Imperial College Business School

Imperial means Intelligent Business

Innovation in Cleantech and Climate Change Solutions

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Imperial College
Business School

Imperial means
Intelligent Business

2. Ideation

Dr. Matthew Lipson
Deputy Director
Undaunted

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Overview of Climate Innovation Sessions

Where are we?

Session 1	Session 2	Session 3	Session 4	Session 5	Deliverables
Problem	Ideation	Discovery	Business Model	Pitching	Pitch pack

Teamwork /Spirit — Work, creativity — Delivery

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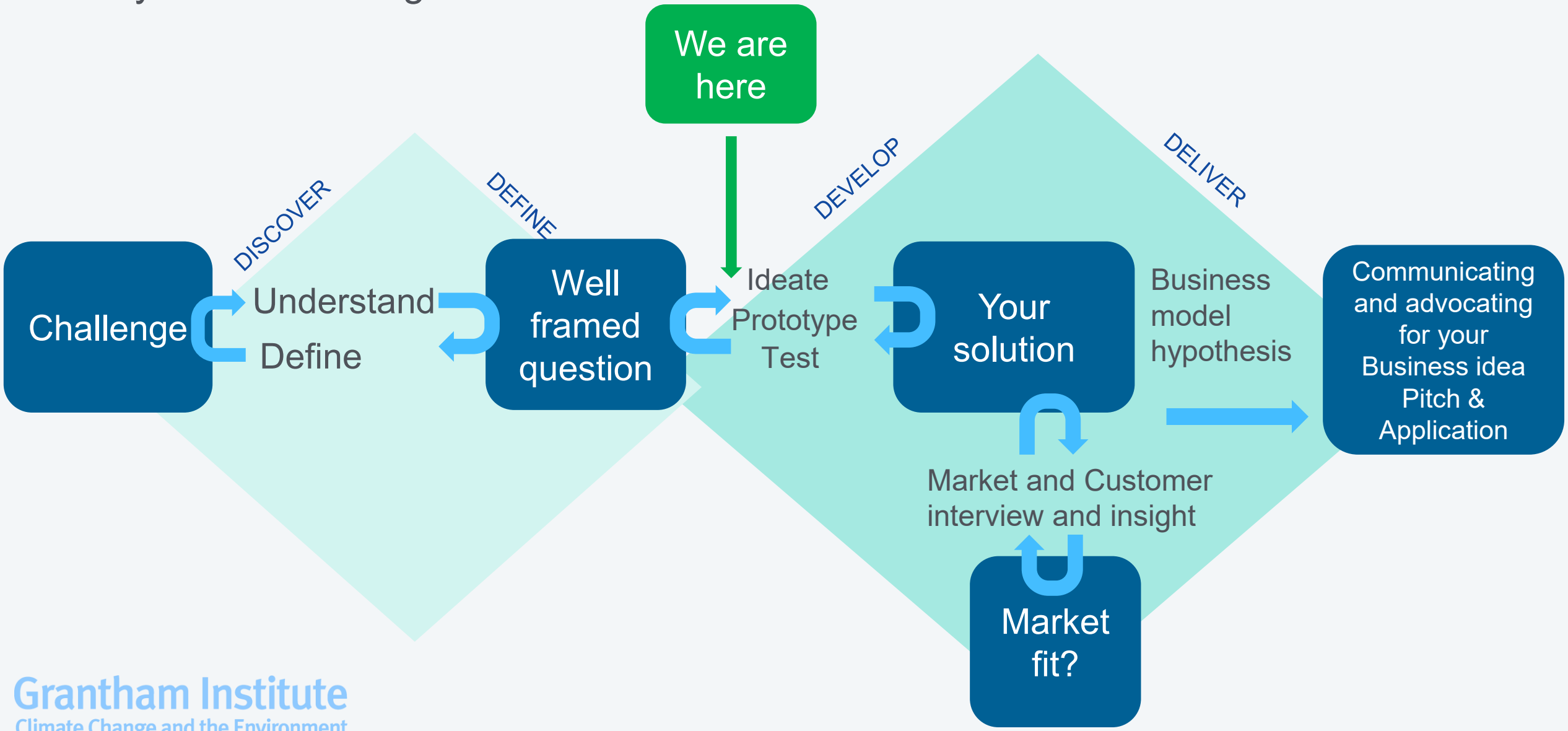
What you will be doing



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Innovation: an example

How could we improve this course?

1: What's the problem

- What's wrong with this course

2: How could you improve it?

- If you were in charge, what would you do?
- If you were all powerful, what would you do?

3: Let's decide!

- If you had to decide today, what would you do?

Climate innovation

Success means building the right ‘it’

Desirable

Customers want 'it'

Commercially viable

‘It’ can pay for itself

Impactful

‘It’ is planet positive

Technically feasible

You can deliver 'it'

Innovation ≠ invention

Climate innovation

Categories of innovation

PROFIT MODEL
The way in which you make money

STRUCTURE
Alignment of your talent and assets

PRODUCT PERFORMANCE
Distinguishing features and functionality

SERVICE
Support and enhancements that surround your offerings

BRAND
Representation of your offerings and business

NETWORK
Connections with others to create value

PROCESS
Signature or superior methods for doing your work

PRODUCT SYSTEM
Complementary products and services

CHANNEL
How your offerings are delivered to customers and users

CUSTOMER ENGAGEMENT
Distinctive interactions you foster

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GET A FREE RAZOR PRODUCTS SHAVING TIPS ABOUT

Gillette

1900

PROTOTYPE

King C. Gillette has the revolutionary idea of disposable blades so thin and so strong they were deemed impossible to forge by MIT-trained scientists. By 1901, he'd proven them wrong with his breakthrough innovation.

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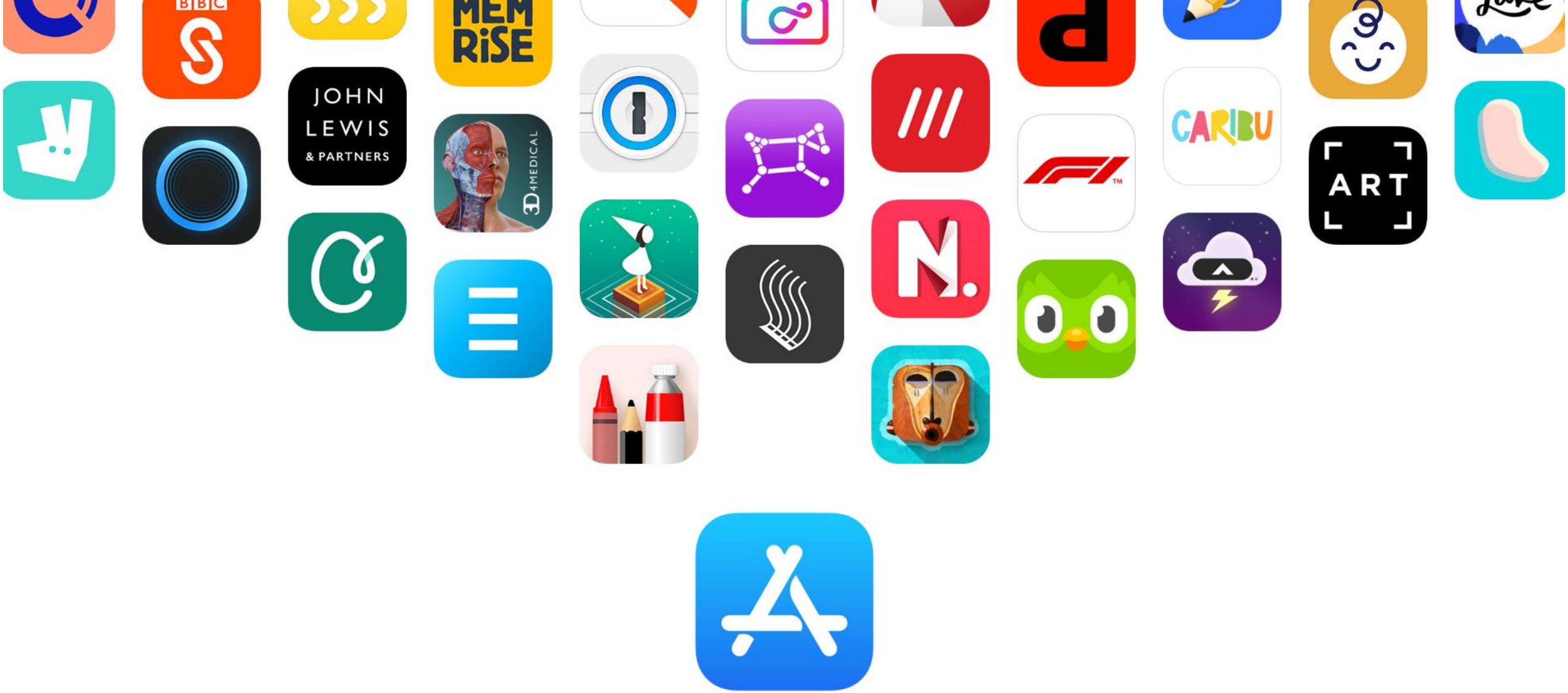

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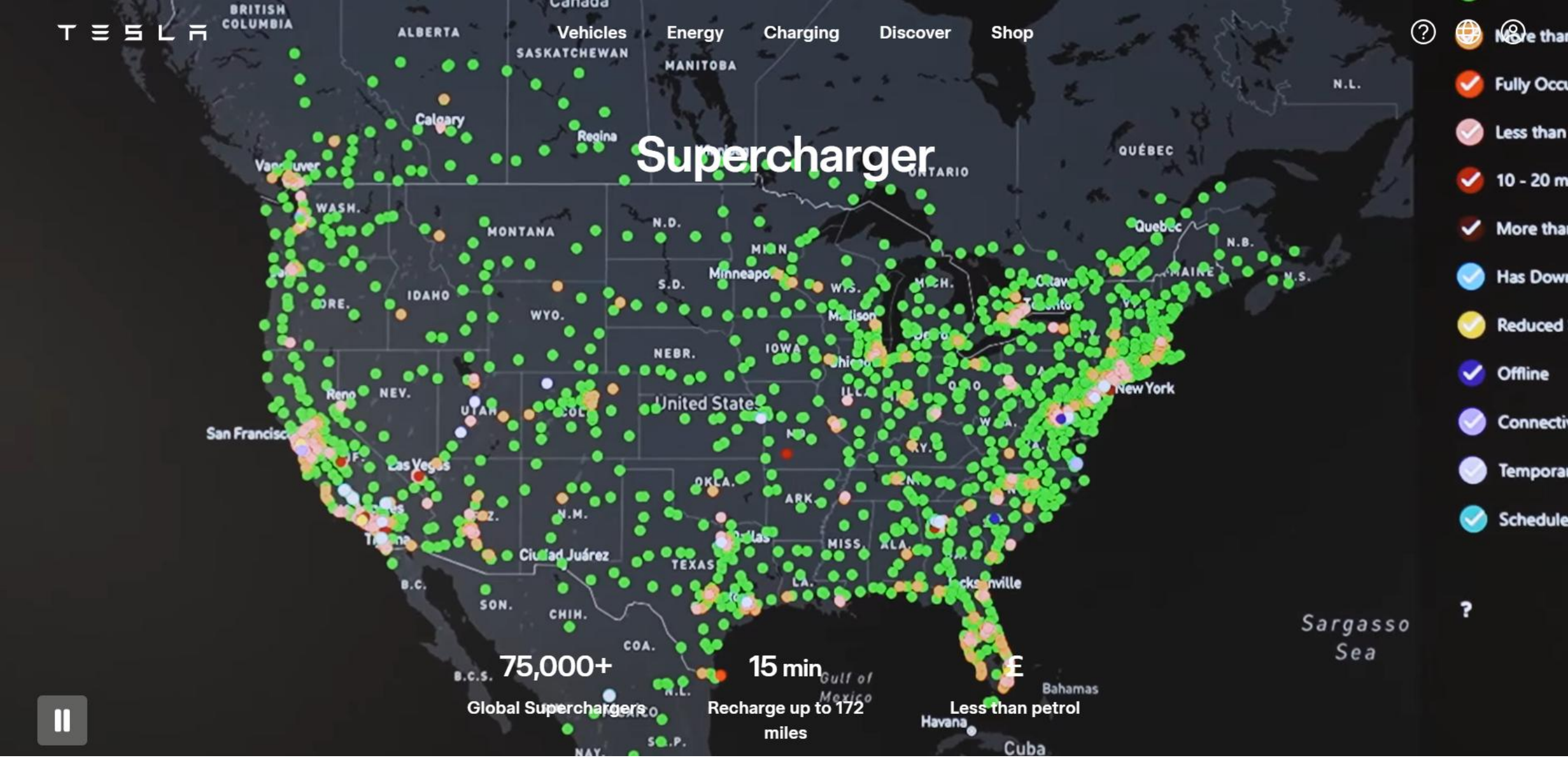
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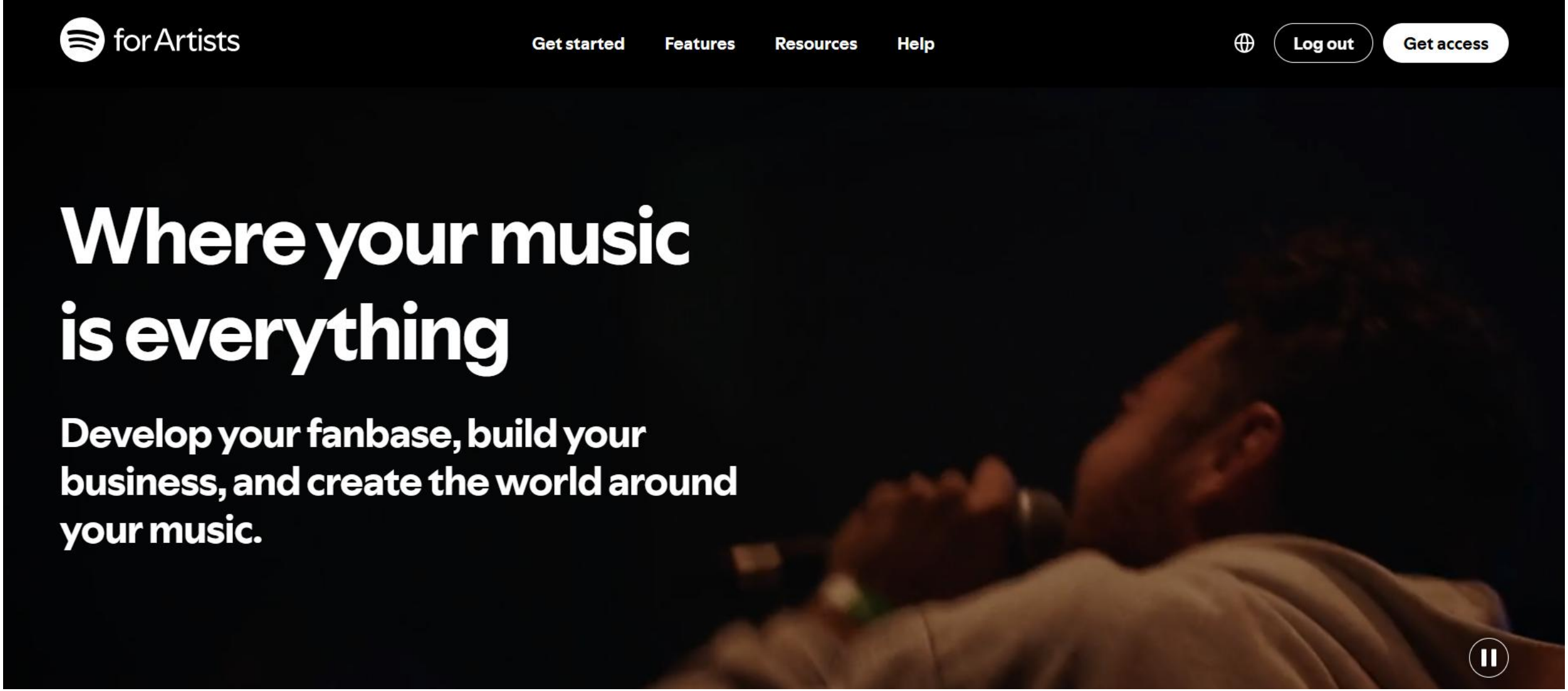
Schedule
75,000+
Global Superchargers
15 min
Recharge up to 172 miles
£
Less than petrol



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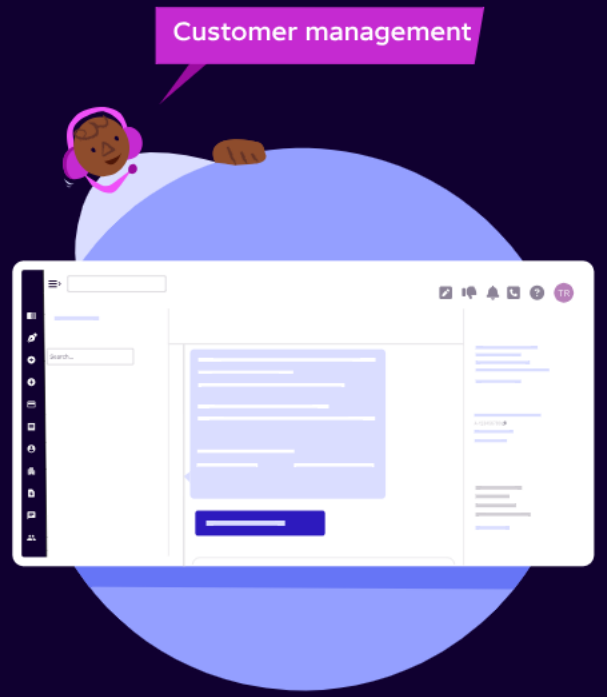

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TOYOTA
MANUFACTURING UK

Home / Lean approach

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About Ørsted

We are a global leader in offshore wind with 10.2 GW installed and over 30 years of experience.

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Climate innovation

Product

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Product system

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Service

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Channel

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Brand

JUST DO IT®

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Customer engagement

duolingo

ECOSIA

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Climate innovation

Categories of innovation

Profit Model	Network	Structure	Process	Product Performance	Product System	Service	Channel	Brand	Customer Engagement
CONFIGURATION				OFFERING		EXPERIENCE			

PROFIT MODEL
The way in which you make money

*For example, how **Netflix** turned the video rental industry on its head by implementing a subscription model*

STRUCTURE
Alignment of your talent and assets

*For example, how **Whole Foods** has built a robust feedback system for internal teams*

PRODUCT PERFORMANCE
Distinguishing features and functionality

*For example, how **OXO Good Grips** cost a premium but its "universal design" has a loyal following*

SERVICE
Support and enhancements that surround your offerings

*For example, how "Deliver WOW through service" is **Zappos'** #1 internal core value*

BRAND
Representation of your offerings and business

*For example, how **Virgin** extends its brand into sectors ranging from soft drinks to space travel*

NETWORK
Connections with others to create value

*For example, how **Target** works with renowned external designers to differentiate itself*

PROCESS
Signature or superior methods for doing your work

*For example, how **Zara's** "fast fashion" strategy moves its clothing from sketch to shelf in record time*

PRODUCT SYSTEM
Complementary products and services

*For example, how **Nike+** parlayed shoes, sensors, apps and devices into a sport lifestyle suite*

CHANNEL
How your offerings are delivered to customers and users

*For example, how **Nespresso** locks in customers with its useful members only club*

CUSTOMER ENGAGEMENT
Distinctive interactions you foster

*For example, how **Wii's** experience draws more from the interactions in the room than on-screen*

Ideation

What works?

1: Unstructured groups underperform

- Can feel fun
- But can be very unproductive

2: Structure can reduce known failures

- Solo then share
- Diverge then converge
- Review to see gaps

3: Insight is unpredictable

- Immerse in the problem
- Search for inspiration

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Brainstorm activity

Three steps to structure

1. Define your focus

- Key insights about the problem to solve or the opportunity to take

2. Diverge

- Private one idea/sticky
- Post stickies silently
- Build-on 3 ‘make it cheaper, better by xxx’
- Lens-switches (pick 3 see next slide)

3. Converge

- Cluster -> vote -> discuss -> select
- Decide how to test your riskiest assumption

Examples of lens switches

Notice how they change your ideas

1. Make it better: simpler, faster, cheaper
2. Reverse assumptions (opposite world)
3. Change what you sell (kwh -> comfort)
4. Swap who does what (warmth on prescription)
5. Analogy (the Uber for heat pump installs)
6. Extremes (design for 10x scale, or 1/10 budget)
7. Combine and remix (fuse any 2, add a service)
8. Dare to dream (magic wand & work backwards)
9. Create a no-brainer (for cautious/anti-person)
10. Define the journey (moments of magic & dropout

Brainstorm activity

Converge and test

1. **Cluster** – silent, then discuss

2. **Vote** – silent, then discuss

3. **Discuss** top 3
   - What's the proposition?
   - What are the pros/cons?
     (impact, desirability, viability, feasibility)

4. **Test** your 'best' one
   - Select one to move forward
   - Find the riskiest assumption
   - Discuss how to test it

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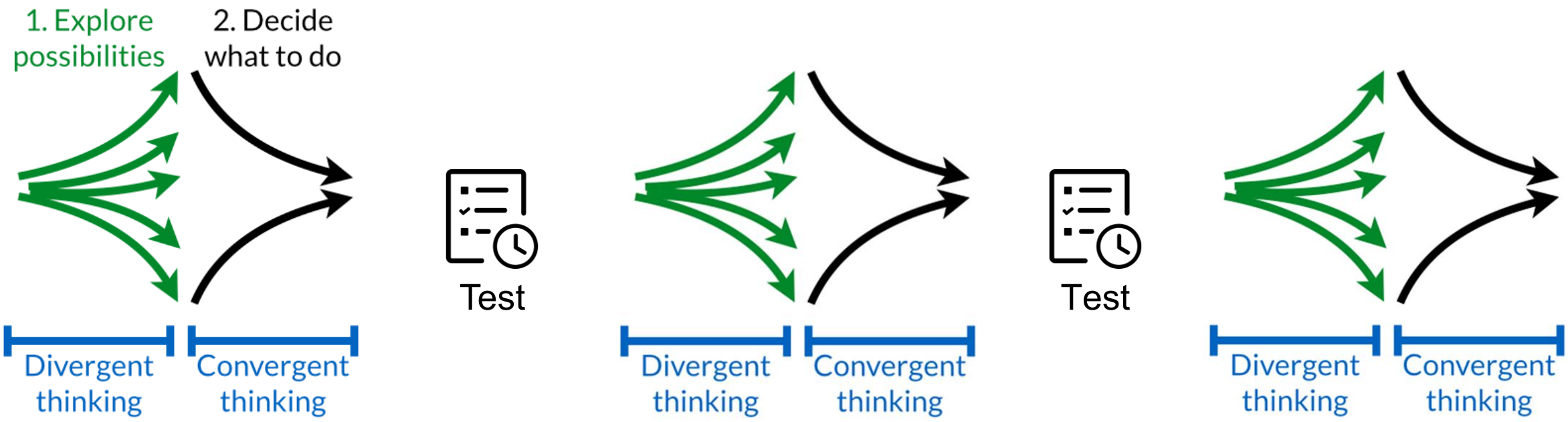
It’s a process

Iterate and focus



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Reflections

What did you notice?

1: Structured vs. unstructured

- Which was more fun?
- Which was more productive?

2: How did the structure help?

- Solo then group
- Think then talk
- Rank to prioritise
- Focus then switch
- Diverge then converge

3: Will your idea work?

- Ideas are cheap
- Failure is expensive

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Pick your idea

By next week

Pick your best idea

- Set a deadline
- Continue to brainstorm
- Choose your idea

Will it work?

- What are the assumptions
- How bad would it be if each was wrong?
- How will you test the riskiest?

Prepare to discover

- Return with an idea to test

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